Functions of a Random Variable

$X \Rightarrow$ indip. var. i.e: credit spread of corp. bond

$Y = g(X) \Rightarrow$ i.e. daily dollar change in portfolio's value

$Y = g(X) = -(\text{Portfolio DV01}) \cdot X$

X Y

PDF: f_X $\qquad$ f_Y

$\downarrow$ $\qquad$ $\uparrow$

CDF: $F_X = P(X < x) \longrightarrow F_Y$

$= \int_{-\infty}^{x} f_X(x)\,dx$

$$X \sim N(\mu, \sigma^2) \Rightarrow Y = aX + b$$

$$F_Y(y) = P(Y < y) = P(aX + b < y) =$$

$$= P\left(X < \frac{y-b}{a}\right) =$$

$$= F_X\left(\frac{y-b}{a}\right)$$

$$f_Y(y) = \frac{d}{dy} F_X\left(\frac{y-b}{a}\right) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right)$$

$$Y \sim N(a\mu + b, a^2\sigma^2)$$

$\Rightarrow$ Chi Squared Distribution

$$X \sim N(0,1)$$

$Y = X^2 \longrightarrow$ Chi - Squared

I.e.: Difference between predictions of my model vs observations in real world should follow N distribution.

& sum of the square of these errors should follow the χ^2 Distribution

$$F_Y(y) = P(Y < y) = P(X^2 < y) = P(-\sqrt{y} < X < \sqrt{y}) =$$

$$= \Phi(\sqrt{y}) - \Phi(-\sqrt{y})$$

REMEMBER $F_X(x) = \Phi(x)$

PDF of Y $\quad f_Y(y) = \frac{d}{dy} F_Y(y) =$

$$= \Phi'(\sqrt{y}) \cdot \tfrac{1}{2} y^{-1/2} - \Phi'(-\sqrt{y}) \cdot \tfrac{1}{2} y^{-1/2} =$$

$$= y^{-1/2} \Phi'(\sqrt{y}) = y^{-1/2} f_X(\sqrt{y})$$

with $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ $\qquad$ $f_Y(y) = \dfrac{y^{-1/2} e^{-y/2}}{\sqrt{2\pi}}$